

WAIFINDERS WORLD

Completion Status Report — v2

Generated: 2026-08-09 · Platform Build: Phase 5 Production Sprint · Status: PILOT-READY

0.8818 ROC-AUC WARN Wildfire Alberta	7.15x Top-10 Lift Open Alberta OGL-A	260 Tests Passing All Modules	8 WARN Domains Active
200 RED-band Bridges NY Multi-Hazard PULSE	58 First Nations Scored BC Wildfire Module	58 Nations at MITIGATION+ BC WARN	17 Dashboard Tabs Live Data

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1. PLATFORM ARCHITECTURE

WAIFINDERS is a multi-hazard, Nation-supporting decision-intelligence platform. All four engines are now PILOT-READY with live outputs. WISE has been upgraded from TRL 3 (design) to TRL 5 (demonstrated in relevant environment) this sprint.

Engine	Function	TRL	Status
WARN	Composite hazard scoring — 8 domains, 0–1 score, 4-tier output	6	PILOT-READY
PULSE	Infrastructure stress — bridges, rail, water mains. Multi-hazard overlay.	6	PILOT-READY
WISE	Multi-hazard decision synthesis. Compound escalation. Ranked actions.	5	PILOT-READY ↑
SENTINEL / S.A.F.E.	Federal-grade audit + BLAKE3-hashed evidence chain.	5	PILOT-READY

2. WARN — ALL 8 ACTIVE HAZARD DOMAINS

Each domain produces a 0–1 WARN score and 4-tier operational state fed into WISE. All 8 are active. BC Wildfire Nations is the newest domain added this sprint.

Domain	Formula Weights	Primary Data Source	Anchor / Top Result
Wildfire — Alberta	FWI 40% · spread 25% · pop 20% · reach 15%	Open Alberta OGL-A 2006-2025	ROC-AUC 0.8818 · Top-10 Lift 7.15x
Wildfire — BC Nations	FWI 35% · proximity 30% · area 20% · spread 15%	BC Wildfire Service public anchor feeds	Kendal-CWFS 0.7246 EMERGENCY (44 km, Donnie Cr
Earthquake	mag 45% · shallowness 35% · pop 20%	USGS ShakeMap / ISC-GEM	Cascadia M9.0 analogue: 0.748 MITIGATION
Tsunami	wave height 45% · mag 35% · reach 20%	NOAA DART buoy network	Tohoku 2011: 0.970 EMERGENCY
Flood Surge	surge 60% · precip 40%	USGS NWIS / NOAA CO-OPS gauges	Sandy 2.74 m: 0.549 MITIGATION
Hurricane	wind 45% · surge 35% · proximity 20%	NOAA NHC HURDAT2 1851–2023	Andrew 1992: 0.814 EMERGENCY
Nuclear	capacity 45% · EPZ pop 35% · power 20%	NRC Power Reactor Status (daily live)	Three Mile Island Point: 0.871 EMERGENCY
BC Wildfire (nations view)	Same as Wildfire-BC — territory centred	BC Wildfire Data Catalogue territory	52 domains; 1 EMERGENCY + 57 MITIGATION (vs 2021+ fi

3. WISE — MULTI-HAZARD DECISION ENGINE (TRL 3 → 5)

WISE was registered in WAIFINDERS_WORLD/06_WISE/ this sprint and upgraded from TRL 3 (design only) to TRL 5 (demonstrated with real hazard data). 06_WISE/ was previously empty.

Domain	Score	Tier
Wildfire	0.0	NORML
Earthquake	0.399	MONIT
Tsunami	0.97	EMERG
Flood Surge	0.549	MITIG
Hurricane	0.814	EMERG
Nuclear	0.871	EMERG

Parameter	Value
WISE Decision	EMERGENCY_RESPONSE
Compound event	True
Elevated domains	6
Elevated list	Earthquake, Tsunami, Flood Surge, Hurricane, Nuclear, Pulse
Logic	base = max tier; +1 if ≥2 elevated; cap EMERGENCY

4. PULSE — MULTI-HAZARD INFRASTRUCTURE SCORING (TRL 5 → 6)

PULSE was upgraded from synthetic wildfire domain only (TRL 5) to full multi-hazard bridge scoring using USGS NSHM 2014, NOAA CO-OPS, and FEMA NFHL authoritative zone data (TRL 6). 04_PULSE/ was previously empty and is now populated.

Component	Detail
Formula	0.65 × poor_condition_risk + 0.35 × age_norm (base); × (1 + 0.25×seismic + 0.20×tsunami + 0.15×flood), capped at 1.0

Component	Detail
Seismic zones	USGS NSHM 2014 (E2014R1) 2%/50yr PGA — NYC metro HIGH (0.17-0.18g)
Tsunami zones	NOAA CO-OPS tide gauge presence + coastal geography
Flood zones	FEMA NFHL SFHA (Zone A/AE) county-level — 11/13 NY counties confirmed
NY bridge results	2,000 bridges: RED=200, AMBER=340, YELLOW=1141, GREEN=319
Water validation	Toronto AUC 0.7384 (n=4,248) · Calgary AUC 0.6626 (n=4,748) — temporal holdout
GATE-004	CLOSED — open-data provenance confirmed for Toronto, Calgary, Kitchener

5. BC WILDFIRE NATIONS — 58 FIRST NATIONS SCORED

New module added this sprint. Scores First Nations with traditional territories in BC fire-risk zones against BC Wildfire Service anchor fires. OCAP data sovereignty principles are embedded — no Nation-specific or cultural site data is collected. Territory centroids use public BC Data Catalogue layers only.

Rank	Nation	Region	Score	Tier	Closest Fire	Dist (km)
1	Kwadacha Nation	Rocky Mountain	0.725	EMERG	Donnie Creek Complex	44.0
2	Tsay Keh Dene Nation	Rocky Mountain	0.667	MITIG	Donnie Creek Complex	82.1
3	Lytton First Nation	Fraser Canyon	0.641	MITIG	Lytton Creek	1.1
4	Halfway River First Nation	Peace	0.61	MITIG	Donnie Creek Complex	120.3
5	Okanagan Indian Band	Okanagan	0.6	MITIG	White Rock Lake	21.4
6	Bonaparte Indian Band	Thompson	0.598	MITIG	Tremont Creek	25.6
7	Nlaka'pamux Nation Tribal Council	Thompson	0.595	MITIG	Lytton Creek	31.6
8	Lower Nicola Indian Band	Nicola	0.593	MITIG	Lytton Creek	33.0
9	Upper Nicola Band	Nicola	0.592	MITIG	White Rock Lake	26.5
10	Okanagan Nation Alliance	Okanagan	0.589	MITIG	White Rock Lake	29.1
11	Westbank First Nation	Okanagan	0.588	MITIG	White Rock Lake	29.6
12	Nooaitch Indian Band	Nicola	0.557	MITIG	Lytton Creek	57.2
13	St'át'imc Nation	St'át'imc	0.555	MITIG	Lytton Creek	58.7
14	Tk'emlúps te Secwépemc	Thompson	0.551	MITIG	Tremont Creek	57.1
15	Kamloops Indian Band	Thompson	0.547	MITIG	Tremont Creek	59.2
16	Coldwater Indian Band	Nicola	0.541	MITIG	Lytton Creek	67.9
17	Splatsin First Nation	Shuswap	0.539	MITIG	White Rock Lake	62.3
18	Canim Lake Band	Cariboo	0.536	MITIG	Horse Lake Complex	44.4
19	Little Shuswap Lake Band	Shuswap	0.53	MITIG	White Rock Lake	68.1
20	N'Quatqua	Sea to Sky	0.527	MITIG	Lytton Creek	76.8

Parameter	Value
Nations scored	58
EMERGENCY_RESPONSE	1
MITIGATION_REQUIRED	57

Parameter	Value
MONITOR / NORMAL	0
Fire data source	failed (HTTP Error 404: Not Found) — using anchor fires — using recent anchor fires (2021+)
Territory source	BC Data Catalogue — public First Nations territory layer (approximate centroids)
Data sovereignty	OCAP principles. No Nation-specific data collected without consent.
Top result	Kwadacha Nation — score 0.7246 EMERGENCY_RESPONSE

6. WILDFIRE VALIDATION — ALBERTA REAL-DATA REPLAY

The primary validated WARN domain. ROC-AUC 0.998 (synthetic, label leakage) is superseded and must not be cited. All current evidence uses real Open Alberta OGL-A data.

Metric	Value	Notes
Dataset	Open Alberta OGL-A 2006-2025	ID: a221e7a0-4f46-4be7-9c5a-e29de9a3447e
Training events	20,848	Years 2006-2019
Test events	6,980	Years 2020-2025 — temporal holdout
High-impact events	158	Large / extreme fire class
ROC-AUC (test)	0.8818	REAL DATA — replaces superseded 0.998
PR-AUC (test)	0.2837	
Top-10 Lift	7.15x	
Top-20 Lift	4.08x	
Mean lead time	274.42 h	
Median lead time	17.75 h	
Geographic scope	Alberta only	Must NOT generalise to BC/Ontario without separate replay
Peer review	NOT COMPLETE	GATE-002 open — external dependency
SUPERSEDED (do not cite)	ROC-AUC 0.998	Label leakage; synthetic dataset (500 events)

7. WAIFINDERS WORLD INTEGRATION — 04_PULSE / 05_WARN / 06_WISE

Three previously empty directories in WAIFINDERS_WORLD are now populated with engine registrations wired to live disaster module outputs. Disaster outputs are staged in outputs/disasters/ (9 JSON files, BLAKE3-hashed).

Directory	Was	Now
04_PULSE/	EMPTY	PULSE_MULTHAZARD_DOMAIN_REGISTRATION.md — bridge formula, USGS/NOAA/FEMA sources, NY resu
05_WARN/	EMPTY	WARN_MULTHAZARD_DOMAIN_REGISTRATION.md — all 8 domains, tier definitions, WISE integration, BLAKE
06_WISE/	EMPTY (TRL 3 'design)	WISE_ENGINE_REGISTRATION.md — live decision state, compound logic, recommended actions, TRL upgraded
outputs/disasters/	NOT PRESENT	9 BLAKE3-hashed JSON outputs: warn_*.json + pulse_*.json + wise_*.json + warn_bc_wildfire_nations.json

8. DASHBOARD — 17 LIVE TABS (dashboard/main.py)

Dashboard upgraded from a shell with placeholder warnings to 17 tabs reading live JSON outputs. Previously dead PULSE, WARN, WISE, and BC Wildfire tabs are now fully wired. Water tab shows GATE-004 CLOSED.

Tab	Was	Now
World Overview	Live	Live — WARN replay metrics, readiness matrix
Doctrine	Live	Unchanged
SENTINEL	Live	Unchanged
PULSE	Wildfire-only stub	LIVE — multi-hazard bridge risk bands, hazard zone sources, priority bridges, AUC validation
WARN	Markdown only	LIVE — 7-domain signal table with tier icons + wildfire replay report
WISE	■ 'Design phase. No outputs available'	LIVE — decision state, elevated domains, recommended actions, cost estimate
RELATE	Placeholder	Unchanged (FRAMEWORK phase)
S.A.F.E.	Live	Unchanged
Wildfire Validation	Live	Unchanged — Alberta replay results
BC Wildfire Nations	NOT PRESENT	NEW — 58 First Nations ranked by WARN score, closest fire, data sovereignty statement
Water Validation	■ 'DESIGN phase. Data agreement not acquired'	GATE-004 CLOSED — AUC results displayed
Digital Twins	■ Concept	Unchanged (CONCEPT phase)
Evidence Ledger	Live	Unchanged
Funding Readiness	Live	Unchanged
Partner Packages	Live	Unchanged
Claim Control	Live	Unchanged

9. GATE CLOSURE STATUS

Gate	Name	Status	Changed This Sprint	Next Action
GATE-001	Official Alberta Wildfire Data	REPLAY_COMPLETE REVIEW_PENDING	No	External peer review
GATE-002	External Peer Review	NOT_CLOSED EXTERNAL	No	Recruit ≥3 reviewers
GATE-003	Production Claim Gate	PILOT_READY PROD_BLOCKED	No	Close 001 + 002
GATE-004	Water Validation Provenance	CLOSED	✓ Confirmed closed	Advance to production data partnership
GATE-005	Bow River Digital Twin	NOT_CLOSED EXTERNAL	No	GIS layers + Treaty 7
GATE-006	Federal SENTINEL Engagement	READY_TO_SEND	No	Send to NRCan + Public Safety
GATE-007	Deployment Operations	CLOSED	No	Review with pilot partner
GATE-008	External Doc Scrub	CLOSED	No	Re-run after doc changes
GATE-009	Tests + BLAKE3 Manifest	CLOSED	260 tests passing	Re-run after code changes

10. READINESS MATRIX — CHANGES THIS SPRINT

Component	Previous Status / TRL	Updated Status / TRL	What Changed
WARN	PILOT-READY TRL 5 Synthetic replay	PILOT-READY TRL 6 Real OGL-A + 7 live domains	6 additional hazard domains live; BC Wildfire Nations added
PULSE	PILOT-READY TRL 5 Synthetic wildfire only	PILOT-READY TRL 6 Real multi-hazard bridge data	USGS/NOAA/FEMA zones; NY 2,000 bridges; 04_PULSE/ populated
WISE	DESIGN TRL 3 No outputs	PILOT-READY TRL 5 Live 7-domain synthesis	Engine built and registered; 06_WISE/ populated; dashboard live
Water Infrastructure	DESIGN TRL 2 GATE-004 open	GATE-004-CLOSED TRL 5 AUC 0.7384/0.6626	Confirmed GATE-004 closed; Toronto+Calgary+Kitchener OGL confirmed
Demo Dashboard	SHELL — dead tabs	LIVE — 17 tabs	PULSE/WARN/WISE/BC wired to live outputs
BC Wildfire Nations	NOT PRESENT	PILOT-READY TRL 5 58 nations scored	New module, new WARN domain

11. APPROVED CLAIMS AND CLAIM BOUNDARY

Approved (with listed qualifiers):

- ✓ WARN is a replay-tested multi-hazard risk-prioritization engine. [Alberta replay: real OGL-A data; external peer review pending; decision-support only]
- ✓ WARN BC Wildfire provides risk scores for First Nations territory proximity using public BC Wildfire Service data. [Territory centroids are approximate; Nations retain all decision authority]
- ✓ PULSE provides multi-hazard operational stress indicators for infrastructure assets. [County-level zone assignment; not a structural engineering assessment]
- ✓ WISE synthesises multi-hazard signals into operational decision states. [Research prototype; not validated for emergency management use]
- ✓ WAIFINDERS SENTINEL is available for pilot and partner discussion.
- ✓ WAIFINDERS supports Nation-controlled decision-making.
- ✓ WAIFINDERS uses BLAKE3 for all evidence hashing — no SHA, no MD5.
- ✓ GATE-004 is closed — water data provenance confirmed for Toronto, Calgary, and Kitchener.

Not approved — must not appear in any external material:

- ✗ WARN is a fully validated forecasting system.
- ✗ WARN predicts wildfires [or any hazard outcome].
- ✗ WARN performance transfers from Alberta to BC or Ontario without a separate replay.
- ✗ BC Wildfire Service endorses WAIFINDERS.
- ✗ WAIFINDERS has used Nation cultural or traditional territory data without consent.
- ✗ PULSE autonomously directs resource dispatch or emergency response.
- ✗ SENTINEL has been adopted by the federal government.
- ✗ Calgary / Toronto / Kitchener endorse WAIFINDERS.
- ✗ [Any citation of] ROC-AUC 0.998 — superseded, label leakage.

12. TEST AND EVIDENCE SUMMARY

Evidence Item	Status	Detail
waifinders_disasters test suite	260 / 260 PASS	7 files: earthquake, tsunamis, flood, hurricane, nuclear, PULSE, WISE, BC Wildfire (51 new)
WAIFINDERS World test suite	73 / 73 PASS	16 files (1 stale-flag — stale flag, not fabrication; GATE-009 CLOSED)
Alberta wildfire replay	ON FILE — BLAKE3 hashed	ROC-AUC 0.8818, PR-AUC 0.2837, Top-10 Lift 7.15x — real OGL-A data
Hurricane WARN output	36,439 bytes	125 Cat-3+ storms; HURDAT2 1851-2023; Andrew 1992 top at 0.814

Evidence Item	Status	Detail
Nuclear WARN output	23,202 bytes	67 US plants; Indian Point 0.871 EMERGENCY; NRC daily status live
Tsunami WARN output	642,943 bytes	Tohoku 2011: 0.970 EMERGENCY; full DART buoy event archive
BC Wildfire Nations output	24,393 bytes	58 nations; Kwadacha 0.7246 EMERGENCY; BLAKE3: 323e97c4...
PULSE bridge output	5,999 bytes	2,000 NY bridges; RED=200; USGS/NOAA/FEMA zones applied
WISE decision output	1,995 bytes	EMERGENCY_RESPONSE; compound=True; 6/7 elevated
BLAKE3 world manifest	CURRENT	175 files; GATE-009 CLOSED
External doc scrub	0 failures	101 docs; GATE-008 CLOSED
Production claim gate	5/11 conditions PASS	PILOT_READY — blocked on GATE-001 (peer review) and GATE-002 (official data)

13. COMPLETED BUILD INVENTORY

Item	Repo	Status
WARN — Wildfire Alberta (14-feature formula)	waifinders_disasters	COMPLETE
WARN — Earthquake (USGS ShakeMap)	waifinders_disasters	COMPLETE
WARN — Tsunami (NOAA DART)	waifinders_disasters	COMPLETE
WARN — Flood Surge (USGS NWIS / CO-OPS)	waifinders_disasters	COMPLETE
WARN — Hurricane (NOAA HURDAT2)	waifinders_disasters	COMPLETE
WARN — Nuclear (NRC daily power status)	waifinders_disasters	COMPLETE
WARN — BC Wildfire Nations (58 First Nations)	waifinders_disasters	COMPLETE
PULSE — Multi-hazard bridge scoring (NY, USGS/NOAA/FEMA)	waifinders_disasters	COMPLETE
WISE — 7-domain decision engine (compound escalation)	waifinders_disasters	COMPLETE
04_PULSE/ domain registration	WAIFINDERS_WORLD	COMPLETE (was empty)
05_WARN/ domain registration (8 domains)	WAIFINDERS_WORLD	COMPLETE (was empty)
06_WISE/ engine registration	WAIFINDERS_WORLD	COMPLETE (was empty, TRL 3→5)
Dashboard — 17 live tabs	WAIFINDERS_WORLD	COMPLETE (was shell)
Readiness matrix — all TRL / status corrections	WAIFINDERS_WORLD	COMPLETE
Disaster outputs staged in outputs/disasters/	WAIFINDERS_WORLD	COMPLETE
WAIFINDERS Foundational Doctrine v2.0	WAIFINDERS_WORLD	COMPLETE
Governance + Indigenous Data Sovereignty frameworks	WAIFINDERS_WORLD	COMPLETE
SENTINEL Audit + S.A.F.E. framework	WAIFINDERS_WORLD	COMPLETE
Alberta wildfire replay validation engine	WAIFINDERS_WORLD	COMPLETE
All gate closure documents (9 gates)	WAIFINDERS_WORLD	COMPLETE
Deployment operations docs (16 documents)	WAIFINDERS_WORLD	COMPLETE
External review package (READY_TO_SEND)	WAIFINDERS_WORLD	COMPLETE
Federal SENTINEL engagement package (READY_TO_SEND)	WAIFINDERS_WORLD	COMPLETE
Metrics / completion status PDF (this document)	waifinders_disasters	COMPLETE

14. REMAINING GAPS AND PRIORITY ACTIONS

Priority	Gap	Blocks	Action
CRITICAL	External peer review (WARN)	'Fully validated' claim	Recruit ≥3 reviewers — REVIEWER_OUTREACH_EMAIL.md ready
CRITICAL	Official Alberta wildland fire data	Production WARN claim	Send OFFICIAL_ALBERTA_WILDFIRE_DATA_REQUEST_LETTER.md
CRITICAL	Federal SENTINEL engagement	Production SENTINEL deployment	Send NRCan + Public Safety Canada package — GATE-006 READY_TO_SEND
HIGH	BC wildfire official data agreement	Production BC WARN claim	Identify BC Wildfire Management Branch contact
HIGH	Nation engagement for BC territory	Nation-specific BC layers	Initiate Nation-to-Nation data agreement process (OCAP)
HIGH	Production water infrastructure plan	Water PULSE enterprise claim	Calgary / Toronto / Kitchener operational data agreement
MEDIUM	Bow River Digital Twin (17 GIS layers)	Digital Twins component	Obtain layers; Treaty 7 governance engagement
MEDIUM	Ontario wildfire validation	Ontario WARN scope	Separate Alberta-equivalent replay — must NOT generalise from Alberta
LOW	RELATE pilot architecture	RELATE pilot	Define pilot scope + dataset
LOW	WISE knowledge corpus	WISE pilot	Domain experts + data curation

Immediate actions (no external dependency):

- Send external peer review invitations — REVIEWER_OUTREACH_EMAIL.md is ready.
- Send Alberta wildfire data request letter — OFFICIAL_ALBERTA_WILDFIRE_DATA_REQUEST_LETTER.md is ready.
- Send Federal SENTINEL package to NRCan (Canadian Forest Service) and Public Safety Canada.
- Identify BC Wildfire Management Branch contact for BC data agreement.
- Update FEDERAL_CONTACT_LOG.csv and OFFICIAL_DATA_ACCESS_TRACKER.csv after each outreach.

CLAIM BOUNDARY — This document is an internal status report. WAIFINDERS is PILOT-READY; enterprise and production claims require GATE-002 (external peer review) and GATE-001 (official data agreement) closure. Wildfire validation: Open Alberta OGL-A 2006-2025, Alberta scope only. BC Wildfire Nations uses public territory centroids only; no Nation-specific data collected. BLAKE3 is the sole permitted hashing standard. All WARN/PULSE/WISE outputs are decision-support tools; human operators retain all authority. Not validated for emergency management use without independent review.

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